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MAST20030: Differential Equations
Tutorial Solutions: Week 6

Solution to Question 1.

(a) Applying $\mathscr{L}$ to both sides gives

$$sY(s) - y(0) + 2Y(s) = \frac{4}{s^2} \;\Rightarrow\; Y(s) = \frac{4}{s^2(s+2)} + \frac{3}{s+2}.$$

The cover method only gives $\dfrac{4}{s^2(s+2)} = \dfrac{2}{s^2} + \dfrac{A}{s} + \dfrac{1}{s+2}$, but substituting $s = 1$ quickly gives $A = -1$. Thus,

$$Y(s) = \frac{2}{s^2} - \frac{1}{s} + \frac{4}{s+2} \;\Rightarrow\; y(t) = 2t - 1 + 4\mathrm{e}^{-2t}.$$

(b) This time, applying $\mathscr{L}$ to both sides gives

$$s^2Y(s) - sy(0) - y'(0) - 2[sY(s) - y(0)] + Y(s) = \frac{1}{s-2}$$
$$\Rightarrow\; Y(s) = \frac{1}{(s-2)(s^2-2s+1)} = \frac{1}{(s-2)(s-1)^2} = \frac{1}{s-2} + \frac{A}{s-1} - \frac{1}{(s-1)^2},$$

again using the cover method. Substituting $s = 0$ now gives $A = -1$, hence

$$y(t) = \mathrm{e}^{2t} - \mathrm{e}^{t} - t\mathrm{e}^{t}.$$

Solution to Question 2.

(a) We can write

$$f(t) = t\,\mathrm{e}^{-t} + (\mathrm{e}^{-t} - t\,\mathrm{e}^{-t})H(t-1).$$

To compute its Laplace transform, we have to remember to make shifts so that each function multiplying $H(t-c)$ is manifestly a function of $t-c$. Thus,

$$f(t) = t\,\mathrm{e}^{-t} - (t-1)\mathrm{e}^{-t}H(t-1) = t\,\mathrm{e}^{-t} - \mathrm{e}^{-1}(t-1)\,\mathrm{e}^{-(t-1)}H(t-1).$$

Since $\mathscr{L}(t\,\mathrm{e}^{-t})(s) = -\dfrac{\mathrm{d}}{\mathrm{d}s}\dfrac{1}{s+1} = \dfrac{1}{(s+1)^2}$, we arrive at

$$F(s) = \frac{1}{(s+1)^2} - \mathrm{e}^{-1}\mathrm{e}^{-s}\frac{1}{(s+1)^2} = \frac{1-\mathrm{e}^{-(s+1)}}{(s+1)^2}.$$

(b) We write the right-hand side as $3 + 2(t-4)H(t-4)$ so that

$$(s^2+1)Y(s) - s = \frac{3}{s} + 2\,\mathrm{e}^{-4s}\frac{1}{s^2}$$
$$\Rightarrow\; Y(s) = \frac{s}{s^2+1} + \frac{3}{s(s^2+1)} + 2\,\mathrm{e}^{-4s}\frac{1}{s^2(s^2+1)}$$
$$= \frac{s}{s^2+1} + \frac{3}{s} - \frac{3s}{s^2+1} + 2\,\mathrm{e}^{-4s}\left(\frac{1}{s^2} - \frac{1}{s^2+1}\right).$$

This is now ready to be inverted:

$$y(t) = -2\cos t + 3 + 2\big(t - 4 - \sin(t-4)\big)H(t-4) = \begin{cases} 3 - 2\cos t, & \text{if } 0 \leqslant t \leqslant 4, \\ 2t - 5 - 2\cos t - 2\sin(t-4) & \text{if } t \geqslant 4. \end{cases}$$

Notice that the solution is *continuous* (and differentiable) at $t = 4$, despite the factor of $H(t-4)$, as discussed in lectures.

Solution to Question 3.

(a) For this initial value problem, taking Laplace transforms gives

$$(s^2+3s+2)Y(s) - s + 1 - 3 = \frac{1}{s} + \mathrm{e}^{-s}$$
$$\Rightarrow\ Y(s) = \frac{1}{s+1} + \frac{1}{s(s+1)(s+2)} + \frac{\mathrm{e}^{-s}}{(s+1)(s+2)} = \frac{1}{2s} + \frac{1}{2(s+2)} + \mathrm{e}^{-s}\left(\frac{1}{s+1} - \frac{1}{s+2}\right)$$
$$\Rightarrow\ y(t) = \frac{1}{2} + \frac{1}{2}\mathrm{e}^{-2t} + \left(\mathrm{e}^{-(t-1)} - \mathrm{e}^{-2(t-1)}\right)H(t-1).$$

Note that the solution is continuous, but not differentiable, at $t = 1$.

(b) This time, taking Laplace transforms gives

$$(4s^2+4s+17)Y(s) = Y(s)\ \Rightarrow\ Y(s) = \frac{1}{4}\frac{1}{(s+\frac{1}{2})^2+2^2}F(s).$$

Since the inverse Laplace transform of $[(s+\frac{1}{2})^2+2^2]^{-1}$ is $\frac{1}{2}\mathrm{e}^{-t/2}\sin(2t)$, we obtain

$$y(t) = \frac{1}{8}\Big(\mathscr{L}^{-1}\left\{\mathscr{L}\big\{\mathrm{e}^{-t/2}\sin(2t)\big\}\mathscr{L}\{f\}\right\}\Big) = \frac{1}{8}\big(\mathrm{e}^{-t/2}\sin(2t)\big) * f(t)$$
$$= \frac{1}{8}\int_0^t \mathrm{e}^{-(t-t')/2}\sin\big(2(t-t')\big)f(t')\,\mathrm{d}t'.$$

(c) First, we note that $\displaystyle\int_0^t y(\tau)\mathrm{d}\tau$ is just the convolution of $y(t)$ with 1, therefore $\mathscr{L}\left\{\displaystyle\int_0^t y(\tau)\mathrm{d}\tau\right\} = Y(s)\mathscr{L}\{1\} = \frac{1}{s}\mathscr{L}\{y\}$. Taking the Laplace transform then gives

$$sY(s) + 4Y(s) + \frac{4}{s}Y(s) = \frac{1}{s} \qquad \Rightarrow \qquad Y(s) = \frac{1}{(s+2)^2}.$$

Applying the s-shift theorem then gives

$$y(t) = \mathscr{L}^{-1}\left\{\frac{1}{(s+2)^2}\right\} = \mathrm{e}^{-2t}\mathscr{L}^{-1}\left\{\frac{1}{s^2}\right\} = t\mathrm{e}^{-2t}.$$